

$$1. y = 4x^3 - 5x^2 + 2x - 7 \quad \text{at } x = 2$$

$$y' = 12x^2 - 10x + 2 \quad 12(2^2) - 10(2) + 2 = \boxed{30}$$

$$2. y = 3x^{-4} - 2x^{-2} + 5 \quad \text{at } x = 1$$

$$y' = \frac{-12}{x^5} + \frac{4}{x^3} \quad \frac{-12}{1^5} + \frac{4}{1^3} = \boxed{-8}$$

$$3. y = 2x^{\frac{5}{2}} - 3x^{\frac{3}{2}} + x \quad \text{at } x = 4$$

$$y' = 5x^{\frac{3}{2}} - \frac{9}{2}x^{\frac{1}{2}} + 1 \quad 5(4^{\frac{3}{2}}) - \frac{9}{2}(4^{\frac{1}{2}}) + 1 = 40 - 9 + 1 = \boxed{32}$$

$$4. y = 2\sin x - 3\cos x + \tan x \quad \text{at } x = \frac{\pi}{4}$$

$$y' = 2\cos x + 3\sin x + \sec^2 x \quad 2\cos\frac{\pi}{4} + 3\sin\frac{\pi}{4} + \sec^2\frac{\pi}{4}$$

$$= 2\left(\frac{\sqrt{2}}{2}\right) + 3\left(\frac{\sqrt{2}}{2}\right) + (\sqrt{2})^2 = \boxed{2 + \frac{5\sqrt{2}}{2}}$$

$$5. y = (x^2 + 3x)(x^3 - 1) \quad \text{at } x = 1$$

$$\text{so } y = x^5 + 3x^4 - x^2 - 3x \quad y' = 5x^4 + 12x^3 - 2x - 3$$

$$y'(1) = 5(1^4) + 12(1^3) - 2(1) - 3 = \boxed{12}$$

$$6. y = (2x - 1)\sin x \quad y' = (2x - 1)\cos x + 2\sin x$$

$$\text{at } x = \frac{\pi}{2} \quad y'\left(\frac{\pi}{2}\right) = (\pi - 1)(0) + 2\sin\frac{\pi}{2} = 2(1) = \boxed{2}$$

$$7. y = \frac{x^2 + 1}{3x - 1} \quad \text{at } x = 1 \quad y' = \frac{(3x - 1)(2x) - (x^2 + 1)(3)}{(3x - 1)^2}$$

$$y'(1) = \frac{(2)(2) - (2)(3)}{2^2} = \boxed{-\frac{1}{2}}$$

$$8. \quad y = \frac{\sin x}{x+1} \text{ at } x=0 \quad y' = \frac{(x+1)(\cos x) - \sin x}{(x+1)^2}$$

$$y'(0) = \frac{(1)(1) - 0}{1^2} = \boxed{1}$$

$$9. \quad y = \frac{x+2}{\cos x} \text{ at } x=0 \quad y' = \frac{(\cos x)(1) - (x+2)(-\sin x)}{\cos^2 x}$$

$$y'(0) = \frac{1 - (2)(0)}{1^2} = \boxed{1}$$

$$10. \quad y = \sin 3x \text{ at } x = \frac{\pi}{9} \quad y' = 3 \cos 3x \quad y'\left(\frac{\pi}{9}\right) = 3 \cos \frac{\pi}{3} = \boxed{\frac{3}{2}}$$

$$11. \quad y = \cos\left(2x - \frac{\pi}{3}\right) \text{ at } x = \frac{\pi}{3} \quad y' = 2(-\sin(2x - \frac{\pi}{3})) \quad y'\left(\frac{\pi}{3}\right) = 2(-\sin \frac{\pi}{3}) = \boxed{-\sqrt{3}}$$

$$12. \quad \text{~~scribble~~} y = \tan 4x \text{ at } x = \frac{\pi}{16} \quad y' = 4 \sec^2 4x$$

$$y'\left(\frac{\pi}{16}\right) = 4 \sec^2 \frac{\pi}{4} = 4(\sqrt{2})^2 = \boxed{8}$$

$$13. \quad y = \sec(2x) \text{ at } x = \frac{\pi}{6} \quad y' = 2 \sec 2x \tan 2x \quad y'\left(\frac{\pi}{6}\right) = 2(2)(\sqrt{3}) = \boxed{4\sqrt{3}}$$

$$14. \quad y = (3x^2 - 1)^4 \text{ at } x = 1 \quad y' = 4(3x^2 - 1)^3(6x) \quad y'(1) = 4(3 - 1)^3(6) = \boxed{192}$$

$$15. \quad y = (2x+1)^{-3} \text{ at } x = 1 \quad y' = -3(2x+1)^{-4}(2) \quad y'(1) = \frac{-6}{3^4} = \boxed{-\frac{2}{27}}$$

$$16. \quad y = \cos\left(\frac{\pi x^2}{3}\right) \text{ at } x = 1 \quad y' = -\sin\left(\frac{\pi x^2}{3}\right) \left[\frac{2\pi x}{3}\right] \quad y'(1) = (-\sin \frac{\pi}{3}) \left(\frac{2\pi}{3}\right)$$

$$= -\frac{\sqrt{3}}{2} \cdot \frac{2\pi}{3} = \boxed{-\frac{\pi\sqrt{3}}{3}}$$

$$17. \quad y = (x^2+1)\cos\left(\frac{\pi x}{2}\right) \text{ at } x = 1 \quad y' = (x^2+1)\left[-\sin \frac{\pi x}{2}\right]\left(\frac{\pi}{2}\right) + 2x \cos \frac{\pi x}{2}$$

$$y'(1) = (2)(-\sin \frac{\pi}{2})\left(\frac{\pi}{2}\right) + 2 \cos \frac{\pi}{2}$$

$$= \boxed{-\pi}$$

$$18. \quad y = \frac{\sin 2x}{x^2+1} \quad \text{at } x=0 \quad y' = \frac{(x^2+1)(2\cos 2x) - 2x \sin 2x}{(x^2+1)^2}$$

$$y'(0) = \frac{(1)(2) - 0}{1^2} = \boxed{2}$$

$$19. \quad y = \frac{x^2+1}{\sin x} \quad \text{at } x = \frac{\pi}{2} \quad y' = \frac{(\sin x)(2x) - (x^2+1)\cos x}{\sin^2 x}$$

$$y'\left(\frac{\pi}{2}\right) = \frac{1(\pi) - 0}{1^2} = \boxed{\pi}$$

$$20. \quad y = (x^2+1)^2(\sin \pi x) \quad \text{at } x = \frac{1}{2}$$

$$y' = (x^2+1)^2(\pi \cos \pi x) + 2(x^2+1)(2x)(\sin \pi x)$$

$$y'\left(\frac{1}{2}\right) = 0 + 2\left(\frac{5}{4}\right)(1)(1) = \boxed{\frac{5}{2}}$$